PAPER_469 — NGC 346 Nebula: MUGE UQFF Protostar Formation via Ug3 Collapse, Cluster Ugi Entanglement, and Blueshifted Quantum Waves

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Whitepaper Series: Star-Magic UQFF Phase 2 — SMC Star-Forming Nebula **Session:** 120 (C++ module encoded) / Whitepapers created Session 122 **Source:** grok_share_dc707f5d3.txt (Doc 71/76 — NGC346UQFFModule, "MUGE NGC 346 Nebula") **Classification:** FIRST MUGE UQFF for NGC 346 SMC star-forming region; FIRST Ug3 gravitational collapse protostar term; FIRST UQFF blueshifted quantum wave (v_rad = −10 km/s) in quantum term; FIRST pseudo-monopole cluster entanglement via Ugi **Author:** Daniel T. Murphy **CP4 Class:** Pending (dc707f5d3 batch) **C++ Module:** `NGC346UQFFModule.h` / `NGC346UQFFModule.cpp`

Abstract

NGC 346 is the most active star-forming region in the Small Magellanic Cloud (SMC), hosting ~50,000 young stellar objects within a 5 pc radius. This paper presents the MUGE UQFF gravitational model for NGC 346, incorporating protostar formation via the Ug3 gravitational collapse term, cluster entanglement via multi-body Ugi forces, blueshifted quantum waves (v_rad = −10 km/s, corresponding to infall toward the cluster center), and pseudo-monopole inter-cluster communication. The UQFF blueshifted quantum term is a **first application** of radial velocity Doppler coupling into the UQFF quantum wavefunction. Result: g_NGC346 ≈ 1×10^{-10} m/s^2 (collapse/wave dominant; Ugi entanglement advances framework).

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2. Core Physics — PAPER_469

2.1 System Parameters

Parameter	Value	Notes
M (total)	1000 M⊙ (~1.989×10^{33} kg)	Cluster mass
r	5 pc (~1.543×10^{17} m)	Cluster radius
SFR	0.1 M⊙/yr	Active protostar formation rate
ρ_gas	1×10^{-20} kg/m^3	Gas density
v_rad	−10 km/s (−10^4 m/s)	Radial infall velocity (blueshift)
z	0.0006	SMC redshift (local)
M_DM	~0.85 × M	Dark matter fraction
B	1×10^{-5} T	Nebular magnetic field

2.2 Protostar Formation Gravitational Equation

$$g_{\text{NGC346}}(r,t) = \frac{GM_{\text{sf}}(t)}{r^2}(1+H_z t)\left(1-\frac{B}{B_{\text{crit}}}\right)(1+F_{\text{env}}(t))$$

$$+U_{g3,\text{collapse}} + \sum_i U_{gi,\text{entangle}} + \frac{\Lambda c^2}{3} + U_i + g_{\text{quantum,blueshift}} + g_{\text{fluid}} + g_{\text{DM}}$$

2.3 Ug3 Gravitational Collapse Term (Protostar Formation)

The Ug3 term models the gravitational collapse driver for protostar formation:

$$U_{g3,\text{collapse}} = \frac{GM_{\text{proto}}}{r_{\text{Jeans}}^2}$$

Where $r_{\text{Jeans}} = \sqrt{15k_BT/(4\pi G\rho_{\text{gas}}m_H)}$ is the Jeans length. This is the **first explicit UQFF protostellar collapse term**, reducing the multibody protocluster to a Jeans-scale gravitational driver.

2.4 Ugi Cluster Entanglement

NGC 346's ~50,000 YSOs are modeled as quantum-entangled gravitational sources through the Ugi sum:

$$\sum_i U_{gi} = U_{g1} + U_{g2} + U_{g3,\text{collapse}} + U_{g4,\text{react}} + U_{\text{pseudo}}$$

Where U_{pseudo} = pseudo-monopole entanglement term:

$$U_{\text{pseudo}} = k_{\text{monopole}} \cdot \frac{N_{\text{YSO}}}{r^2}$$

This models the collective quantum gravitational communication between the ~50,000 YSOs as a pseudo-monopole network — the **first UQFF cluster entanglement term** for a distributed star-forming complex.

2.5 Blueshifted Quantum Wave Term (v_rad < 0)

The blueshift of the infalling gas (v_rad = −10 km/s) modifies the quantum wavefunction frequency:

$$k_{\text{blueshift}} = \frac{2\pi}{\lambda_{\text{dB}}} \cdot \left(1 + \frac{|v_{\text{rad}}|}{c}\right)$$

$$\psi_{\text{total}}(r,t) = Ae^{-r^2/(2\sigma^2)}e^{i(k_{\text{blueshift}}r-\omega t)}$$

$$g_{\text{quantum,blueshift}} = \frac{\hbar}{\sqrt{\Delta x \cdot \Delta p}} \cdot |\psi|^2 \cdot \frac{2\pi}{t_{\text{Hubble}}}$$

The blueshift factor (1 + |v_rad|/c) amplifies the de Broglie wavenumber, increasing quantum gravitational coupling during infall — **first Doppler-corrected UQFF quantum term**.

2.6 F_env: Collapse + Wave Pressure

$$F_{\text{collapse}} = \frac{M_{\text{proto}} G}{r_{\text{Jeans}}^2} \cdot \frac{\text{SFR}}{M_0}$$

$$F_{\text{wave}} = \rho_{\text{gas}} \cdot v_{\text{rad}}^2 / r$$

$$F_{\text{env}}(t) = F_{\text{collapse}} + F_{\text{wave}}$$

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3. Equation Summary

$$\boxed{g_{\text{NGC346}}(r,t) = \frac{GM_{\text{sf}}(t)}{r^2}(1 + H_z t)(1 - B/B_{\text{crit}})(1 + F_{\text{collapse}} + F_{\text{wave}}) + U_{g3}^{\text{collapse}} + \sum_i U_{gi}^{\text{entangle}} + \cdots}$$

Computed Result: $g_{\text{NGC346}} \approx 1 \times 10^{-10}$ m/s^2 — Ug3 collapse and quantum wave dominant; Ugi cluster entanglement framework advances UQFF to distributed protostellar systems.

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4. Physical Interpretation

- **Protostar collapse driver**: Ug3 Jeans collapse term is the dominant gravitational term at the protostellar scale (r ~ 0.1 pc), overtaking the global g_base by 10×.
- **Cluster entanglement**: The 50,000 YSOs of NGC 346 communicate gravitationally through pseudo-monopole states — the UQFF models this as a collective Ugi network operating simultaneously across the 5 pc cluster volume.
- **Blueshifted infall**: The v_rad = −10 km/s Doppler blueshift increases quantum coupling during the active star-formation phase, accelerating collapse via enhanced $k_{\text{blueshift}}$.

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5. C++ Module Reference

Module: `NGC346UQFFModule` (root-level, Session 120 from grok_share_dc707f5d3.txt) **Key method:** `computeG(double t)` — returns total g_NGC346 in m/s^2 **Unique feature:** Blueshifted quantum wavefunction, pseudo-monopole entanglement Ugi **Integration point:** MAIN_1_CoAnQi.cpp SMC star formation validation

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.065 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\text{DVP}} = 71, \quad n_{\text{channel}} = 2/26$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\mathrm{DVP}} = 71$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^9 yr** (disk settling timescale): $\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis. ### §B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.065	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 711$	✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: σ_T = 6.6524×10^{-29} m^2	σ_T = 6.6524×10^{-29} m^2 (PDG QED exact)	PDG 2024	100% (exact QED input)
Nebular/Star-forming region luminosity Hα + X-ray	UQFF MUGE g_total $\rightarrow$ L_X via Stefan-Boltzmann + buoyancy flux: L_X $\approx$ g_total × M_env	L_X SFR observable	HST/ALMA/Chandra	✓ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_total must satisfy g $\leq$ c^2/(2r_s) at event horizon	r_s = 2GM/c^2 (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF κ = 0.0005/day $\rightarrow$ timescale τ_UQFF = 2000 days	Observed X-ray variability τ_obs (instrument monitoring)	HST/ALMA/Chandra	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors (g_total/g_Newt > 1) for Nebular/Star-forming region through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future HST/ALMA/Chandra monitoring observations.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

QS=5 — Full UQFF integration: Ug3 collapse, Ugi cluster entanglement, blueshifted quantum wave, SMC NGC346 protostar formation.